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The impact of singlehood on carbon emissions: Empirical evidence from China☆

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ABSTRACT

Against the backdrop of increasing consumption-based carbon emissions, mitigating household carbon emissions is crucial for achieving China's carbon reduction goals. This study investigates the impact of the household singlehood rate on per capita carbon emissions from a consumption-based perspective. Utilizing panel data from the China Family Panel Studies (CFPS) encompassing 63,412 households from 2010 to 2022, our analysis demonstrates that a rising singlehood rate significantly increases per capita carbon emissions, with a baseline impact coefficient of 0.244. The impact is channeled through three mechanisms: diminished economies of scale in household public goods, insufficient sharing of domestic energy, and increased per capita consumption driven by convenience-seeking behaviors. Heterogeneity analyses uncover systematic disparities: the effect is strongest for males, rural dwellers, the youth, high-income earners and high-education singlehood households. Based on these insights, we propose targeted policy recommendations focused on leveraging learning curves, intervening at sensitive points, and implementing personal carbon budgets to systematically guide and incentivize low-carbon transitions among the singlehood population.

1. Introduction

On January 20, 2025, U.S. President Donald Trump announced the United States' withdrawal from the Paris Agreement and implemented policies to reduce investment in clean energy in favor of traditional fossil fuels—a move expected to adversely affect global carbon emissions (US News, 2025). In March 2025, the International Energy Agency (IEA) released its Global Energy Review 2025, reporting that global energy-related carbon emissions increased by 0.8 % year-on-year in 2024, reaching 37.6 billion tons of CO_2. Atmospheric CO_2 concentrations rose to 422.5 ppm, approximately 3ppm higher than in 2023 and 50 % higher than pre-industrial levels. China's carbon emissions in 2024 remained largely unchanged from the previous year, at 12.6 billion tons, accounting for one-third of global emissions—a share comparable to China's portion of global manufacturing (International Energy Agency (IEA), 2025).

As the world's largest carbon emitter, China has committed to peaking CO_2 emissions by 2030 and achieving carbon neutrality by 2060—objectives collectively known as the “dual-carbon goals” (Ministry of Foreign Affairs of the People's Republic of China, 2020). This strategy necessitates reductions in both lifestyle and production-related emissions. According to the Global Energy Review 2025, weakened real estate markets and reduced infrastructure investment in China led to a 10 % decline in cement production and a more than 5 % reduction in industrial process emissions (International Energy Agency (IEA), 2025). While energy-related carbon emissions in China increased by 0.4 %, industrial process-related emissions decreased by 5 %, indicating a rise in lifestyle carbon emissions. Therefore, mitigating household carbon emissions is crucial for China to achieve its emission reduction targets amid declining industrial emissions.

These macro-trends raise a critical question: what factors are driving the growth of carbon emissions in China? The determinants of carbon emissions are multifaceted, encompassing economic, demographic, and policy factors at the macro level; industrial and energy structure factors at the meso level; and technological innovation and individual behavior at the micro level. Economic growth typically correlates with increased energy consumption and higher emissions, with China's rapid industrialization being a major driver (Cui et al., 2023). Population size directly

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affects total energy demand, while demographic shifts such as urbanization and aging (Guo et al., 2023) also influence emission patterns. Government policies, including carbon emission trading systems (ETS) (Yu et al., 2023) and building emission standards (Du et al., 2024), significantly impact carbon emissions. Heavy industries are typically the largest emitters (Cui et al., 2023), and fossil fuels (coal, oil, natural gas) remain primary sources of carbon emissions, whereas renewable energy (solar, wind, hydro) offers low or zero emissions (Temiz and Dincer, 2023). Technological advances in energy efficiency, new energy systems, carbon capture and storage (CCS), and other innovations play a fundamental role in reducing carbon intensity (Na et al., 2024).

At the micro level, individual consumption patterns and household structure characteristics will influence per capita carbon emissions.Individual consumption habits—such as travel modes and dietary patterns (Khan et al., 2024)—directly affect carbon emissions. Behaviors like using public transportation, reducing meat consumption, and purchasing energy-efficient appliances can lower one's carbon footprint. Household structure is closely linked to consumption patterns and indirectly affects carbon emissions. Variations in household characteristics—including number (household effect), distribution (distribution effect), size (scale effect), per capita energy use (consumption effect), and energy choices (choice effect)—all influence emission intensity. Contrary to scale and distribution effects, the number of households has a positive and significant impact on CO_2 emissions (Shigetomi et al., 2018).

Regarding marital structure, consumption patterns of single individuals differ markedly from those of non-single groups. Data from South Korea suggest that an increase in single-person households in metropolitan areas reduces per capita greenhouse gas (GHG) emissions, partly due to smaller dwelling sizes and more efficient use of shared living spaces (Cho et al., 2021). In contrast, Japanese data indicate that single-person households have a larger carbon footprint than non-single households, particularly among single women, mainly due to higher energy use and spending on clothing and healthcare (Huang et al., 2024). This contradiction between Korean and Japanese findings presents a paradox: does singlehood increase or decrease per capita carbon emissions? The academic community remains divided. Chinese data suggest that a higher proportion of single households leads to reduced electricity consumption and thus lower carbon emissions (Wang et al., 2023). However, this analysis considers only direct emissions from electricity use, underestimating non-electric energy use and indirect emissions linked to consumption.

Currently, few studies explore how China's singlehood rate indirectly affects per capita carbon emissions through household consumption. Using CFPS household survey data from 2010 to –2022, this study analyzes the impact of singlehood on per capita carbon emissions from a consumption perspective, offering theoretical insights into how household structure influences carbon emissions within the context of the second demographic transition.

While existing literature has largely focused on developed economies, a significant research gap remains regarding the impact of singlehood on carbon emissions in developing countries—especially given structural differences in energy systems and consumption behaviors. This study contributes a unique empirical analysis from China, the world's largest carbon emitter and largest developing economy, to elucidate how rising singlehood influences emission trajectories. Its innovations are twofold: first, it pioneers the measurement of singlehood's impact on per capita carbon emissions in China; second, it investigates the mechanisms—through economies of scale in high-value assets, energy sharing, and convenience-driven consumption—by which singlehood affects emissions.

2. Literature review and hypothesis proposed

Global climate change has become an increasingly pressing issue, with countries actively seeking pathways to achieve carbon neutrality goals (Xuan et al., 2025) (Nurgazina et al., 2023). Against this backdrop, in-depth research into the drivers of carbon emissions has become particularly crucial. Traditional carbon emission analyses have primarily focused on macro-level factors such as industrial structure, energy consumption, technological progress, and policy impacts (Oyewo et al., 2024) (Cui and Cao, 2024) (Hu et al., 2024) (Sun et al., 2023) (Li et al., 2024).

However, in recent years, the impact of demographic shifts on carbon emissions has gradually drawn academic attention. Population aging is one of the key factors influencing household carbon emissions. Panel data from 30 Chinese provinces spanning 2003 to 2019 reveal that population aging significantly impacts energy-related carbon emissions from daily life among both urban and rural residents (Yuan et al., 2024). Additionally, urbanization influences carbon emissions by affecting economic production activities, energy demand, and lifestyle and consumption behaviors (Ma and Ogata, 2024). Among these, the rising rate of singlehood—as a significant social phenomenon—and its potential influence on carbon emissions have emerged as a new research focus. The rising rate of single individuals is closely linked to urbanization and demographic shifts, potentially influencing carbon emissions through similar or even unique mechanisms.

Although current literature directly addressing "the impact of single-person households on carbon emissions" is limited, several potential mechanisms can be inferred from related studies.

Inefficient residential energy use.Single-person households typically reside in relatively independent living units. Even if their living space is smaller than the average for multi-person households, the per capita living space is often larger. This implies that each single individual may consume more residential energy (e.g., heating, cooling, lighting) than the per capita consumption of members in multi-person households. For instance, whether an apartment is occupied by one or two people, the heat loss area of its exterior walls and the energy consumption of major appliances like refrigerators may not differ significantly, leading to higher per capita energy consumption when occupied by a single person. Carbon emissions during the production phase of building materials are a critical component of the structure's full life cycle. Increased window-to-wall ratios (WER) may elevate a building's embodied carbon emissions. If single-person households prefer residences with expansive windows or greater independence, their dwellings may exhibit a relatively higher carbon footprint.

Differences in consumption patterns.The consumption behaviors of single individuals may differ from those of household units. For example, singles may be more inclined to purchase smaller-packaged goods, potentially leading to increased packaging material consumption and waste generation, thereby raising carbon emissions during production and disposal processes. Additionally, singles may dine out more frequently, and the carbon intensity of the food service industry is significant, encompassing multiple stages such as ingredient production, processing, transportation, and energy consumption in dining establishments. Shifts in food consumption patterns, such as adopting more plant-based diets, can help reduce the environmental impact of food consumption. However, if the consumption patterns of single individuals do not evolve toward greater sustainability, they may instead increase carbon emissions.

Transportation Patterns.Single individuals may be more inclined to use private transportation, particularly in urban areas. Due to the absence of shared travel needs among family members, their willingness to carpool or use public transportation may be lower. The adoption of electric vehicles (EVs) represents a significant option for reducing transportation greenhouse gas emissions. While the emissions impact of EV charging times requires further understanding, the widespread use of private gasoline-powered vehicles undoubtedly increases carbon emissions. Rail transportation, particularly high-speed rail (HSR), holds the potential for lower carbon emissions over its lifecycle. If singles' travel patterns fail to fully leverage these low-carbon transportation options, their carbon footprint will further increase.

2.1. Declining scale effects of high-value shared household goods lead to increased per capita carbon emissions

The rise in singlehood directly leads to an increase in the number of households and the number of people living alone, and an increase in the per capita housing area and the number of dwellings per capita, leading to a decline in the scale effect and sharing of household consumer goods and energy. First, energy resources (e.g., heating and cooling) traditionally shared by multiple-person households are dispersed into more individual units, and household scale effects decline, with single-family households consuming 1.5–2 times more energy per unit area than multiple-person households (Cho et al., 2021). Second, the need for single-person households to independently purchase appliances such as refrigerators and washing machines leads to an increase in energy consumption per unit of service (Won et al., 2014). Third, smaller homes favored by single-person households have higher surface area-to-volume ratios and lower efficiency for winter heating and summer cooling (Schubert et al., 2012). Fourth, single-person households have lower car-sharing rates, higher vacancy rates, and increased carbon emissions from gasoline.

2.2. Carbon-intensive lifestyles of singlehood with a preference for convenient living lead to increased per capita carbon emissions

The singlehood group's consumption preference is characterized by “immediacy, convenience and personalization”, and the increase in per capita consumption leads to more carbon footprints. First, the high frequency of eating out has led to a surge in takeaway packaging waste, and Chinese singletons' takeaway orders are higher than those of home users, resulting in an increase in carbon emissions from plastic waste (Long et al., 2022). Second, the life cycle compression of carbon emissions as clothing and electronics are updated more frequently (Tan-et al., 2024). Third, the higher frequency of online purchases and higher use of cell phones in singlehood households raises the carbon emissions related to logistics and network signaling.

2.3. Low sharing of household living energy leads to higher per capita carbon emissions

The low degree of sharing of energy consumption among the singlehood group in the three areas of kitchen fuel, winter heating and home appliance use leads to higher per capita carbon emissions.

2.4. Research hypothesis

Accordingly, the following hypotheses are proposed:

Hypothesis 1. The singlehood rate has a positive effect on per capita carbon emissions.

Hypothesis 2. The singlehood rate has a positive impact on per capita carbon emissions through three major paths: reducing the household size effect of high-value assets, improving consumption convenience, and reducing the sharing of living energy.

3. Materials and methods

3.1. Modeling

$$\ln(\text{percarbon}_{it}) = \beta_0 + \beta_1 \text{singlhood}_{it} + \sum_{k=1}^{K} \gamma_k \text{control}_{k,it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

The variance inflation factors for all variables are less than 5, indicating no significant multicollinearity. The Hausman test indicates that a fixed-effects model is required.This study employs a district-time dual fixed-effects panel model (Formula 1) with the following specifications:

The dependent variable is the natural logarithm of per capita carbon emissions ($\ln(\text{percarbon}_{ct})$), representing carbon intensity geography. The core explanatory variable is the household singlehood rate (singlhood_{ct}), defined as the proportion of single adult individuals within a household relative to the total household size. Control variables comprise K socioeconomic indicators ($X_{k,ct}$; see Table 1), with β_0 denoting baseline emissions, β_1 quantifying the net marginal effect of singlehood on emissions, and γ_k capturing control variable moderations

Triple rationale for county-level dual fixed effects:

County fixed effects absorb time-invariant district characteristics, eliminating provincial-scale l aggregation bias.

Temporal fixed effects capture nationwide initiatives, while county effects filter localized interventions.

Leveraging intra-district temporal variation within our high-dimensional panel satisfies strict exogeneity.

The stochastic disturbance term ε_{ct} follows i.i.d. normality $(\varepsilon_{ct} \sim N(0, \sigma^2))$.

3.2. Methods

3.2.1. Methods introduction

In the field of calculating household consumption carbon emissions, a substantial body of literature has emerged in recent years, primarily encompassing input-output methods and life-cycle assessment approaches. Multi-Regional Input-Output (MRIO) models have been widely used to track indirect carbon emissions from household consumption. Zhang et al. (2023) constructed a multi-regional input-output database in China and combined it with micro household survey data to reveal the decoupling mechanism of carbon emissions from urban and rural household consumption (Zhang et al., 2023a); for the specificity of the urban scale, Long et al. (2019) proposes an improved IO model based on purchaser price, which solves the problem of traditional producer

Table 1
Household consumption categories and corresponding industrial production sectors.

Consumption Categories	Industrial Sector Categories
Food	Agricultural and Sideline Food Processing, Food Manufacturing, Beverage Manufacturing, Tobacco Industry
Clothing	Textile Industry; Textile Apparel, Footwear, and Headwear Manufacturing; Leather, Fur, Feather (Down), and Related Products Industry
Housing	Construction; Production and Supply of Electric Power and Heat; Production and Supply of Gas; Production and Supply of Water
Household Goods and Services	Wood Processing and Wood, Bamboo, Rattan, Palm, and Straw Products; Furniture Manufacturing; Electrical Machinery and Equipment Manufacturing; Metal Products; Plastic Products Industry
Transportation, Communications, and Postal Services	Manufacture of Transportation Equipment; Manufacture of Communication Equipment, Computers, and Other Electronic Devices; Transportation Services; Postal and Savings Services
Education, Culture, and Entertainment	Education, Culture, and Entertainment; Manufacture of Paper and Paper Products; Manufacture of Cultural, Educational, and Sports Goods; Printing and Reproduction of Recorded Media; Manufacture of Instruments and Meters; Manufacture of Cultural and Office Machinery
Healthcare	Pharmaceutical Manufacturing
Other Goods and Services	Wholesale and Retail Trade; Accommodation and Food Services; Handicrafts and Other Manufacturing

Source: Compiled according to the National Economic Industry Classification (GB/T4754-2017) released by the National Bureau of Statistics of China in 2017 price model's biased allocation of carbon emissions in the trading chain (Long et al., 2019); Feng et al. (2021) using a nested global multiregional IO model to analyze U.S. household carbon emissions and found that the top 10 % income group emits 3.8 times more carbon than the bottom 50 % group, with luxury consumption and international air travel being the main contributing sources (Feng et al., 2021). The life cycle assessment (LCA) approach has become popular in recent years. Huang et al. (2024) used the EIO-LCA method to quantify the full life-cycle carbon emissions of Japanese households, and found that the implied carbon emissions of high-income households were significantly higher than those of low-income groups in the transportation and education categories of services (Huang et al., 2024); Khan et al. (2024) used life cycle assessment to quantify household food carbon emissions in China and Europe, and found that rural households in China have a significantly higher share of carbon emissions from food transportation (33.7 %) than in Germany (18.2 %) due to the low coverage of the cold chain, but the carbon intensity of the processing process is only 1/3 of that in Europe (Khan et al., 2024).

The aforementioned studies indicate that while life cycle assessment (LCA) can reveal detailed emissions across the entire consumption chain, its high data intensity requirements and insufficient grassroots statistical coverage—particularly in the informal economic sectors of developing countries—constrain the availability and spatio-temporal continuity of full life cycle data.

Given the systemic bottlenecks in obtaining full-cycle data at the consumption end using LCA methods, this study shifts to an application framework based on the input-output approach. Specifically, carbon emission intensity calculations are derived from the product of energy intensity and standard coal emission coefficients.

3.2.2. Calculation process

The core logic of this method lies in linking household consumption expenditures with sectoral carbon emission accounts, thereby circumventing LCA's reliance on micro-process data. Its implementation involves two major steps:

Consumption-Industry Mapping: Establishing correspondences between eight major categories of household consumption expenditures and 42 related industrial sectors based on the National Economic Industry Classification (GB/T 4754-2017) (see Table 1);

Emission Factor Calibration:Calculating carbon emission intensity coefficients for each consumption category based on the industry mapping relationships. The carbon emission intensity is calculated by multiplying the energy intensity with the carbon emission coefficient of standard coal, and because of the limitation of the availability of whole-life consumption data of the life-cycle approach, this paper finally chooses the input-output approach. The application of the input-output method involves correlating various types of consumption expenditures with the carbon emissions of the corresponding industries. Before carrying out the calculation of the carbon emission coefficients of various types of household consumption, the first task is to clarify the close connection between the eight types of household consumption expenditure and the corresponding industrial sectors.

The classification of household consumption expenditures is based on the following:

This study adopts a classification of household consumption expenditures into eight categories: food, tobacco, and alcohol; clothing; housing; household goods and services; transportation and communication; education, culture, and entertainment; health care; and other goods and services. This classification rigorously follows the official “Classification Standard for Chinese Household Consumption Expenditures (2013)” (Announcement No. 22 of the National Bureau of Statistics). The framework offers three principal advantages:

Comprehensive Coverage: These eight categories encompass fundamental household needs such as food, clothing, housing, and transportation. Empirical assessments confirm that this classification captures most of household carbon emissions, as validated using the China Emissions Accounts and Datasets (CEADs) 2023 edition.

Data Traceability: Each consumption category can be accurately traced to 42 corresponding industrial sectors according to the National Economic Industry Classification (GB/T 4754-2017), providing a robust foundation for input-output analysis.

International Comparability: The classification is fully consistent with the consumption categories defined by the United Nations System of Environmental-Economic Accounting (SEEA), enabling reliable cross-country comparative research.

The standard coal emission factor, as a widely used parameter for estimating carbon emissions, offers the core advantage of uniformly converting various forms of energy consumption into standard coal equivalents. This provides a standardized, comparable benchmark for carbon emission calculations at the macro level. The primary reasons for adopting this factor in this study are data availability and computational consistency: standard coal consumption is a commonly reported indicator in provincial and national energy statistics, facilitating alignment with official energy balance sheets and macroeconomic data.

Non-fossil energy sources (such as nuclear power, hydropower, wind power, solar power, etc.) do indeed have corresponding standard coal conversion factors in energy statistics. These factors are primarily used to measure the energy value contained within these sources, enabling unified statistical tracking and comparison with fossil fuels. Although non-fossil energy sources have lower carbon emissions, calculating their specific carbon footprint requires household energy consumption data and the application of carbon emission life cycle assessment methodologies. This represents a promising avenue for future research.

The carbon emission factor of standard coal represents the level of carbon emissions per unit of energy. A more accurate measurement can be achieved by multiplying this factor with the energy intensity corresponding to a specific consumption category. Owing to variations in energy consumption across different industries, the energy intensity for each consumption category is derived from the ratio of standard coal consumption to the total output value of all relevant industries associated with that category. The household carbon emissions are calculated as follows:

$$CI_i = w \times \sum_{n=1}^{N} Ind_{n,i} \Big/ \sum_{n=1}^{N} \Delta_{n,i} \tag{2}$$

$$CE = \sum_{i=1}^{8} (CS_i CI_i) \tag{3}$$

Equation (2) establishes the emission coefficient at the sectoral level by converting industrial energy inputs into carbon intensity per unit of consumption expenditure, whereas Equation (3) applies these coefficients to household-level expenditure data to quantify actual emissions. Specifically, CI denotes the carbon intensity of consumption category i, measured in kilograms of carbon dioxide per yuan ($kgCO_2$/yuan); w represents the carbon emission coefficient of standard coal, taking the value of 2.493 based on standard coal conversion; Ind indicates the standard coal consumption of industry n within consumption category i, measured in kilograms of coal equivalent (kgce); Δ refers to

Table 2
Carbon emission factors corresponding to eight types of household consumption.

Consumption Categories	Industrial Sector Categories
Food	0.418
Clothing	0.561
Housing	0.708
Household Goods and Services	0.484
Transportation, Communications, and Postal Services	0.827
Education, Culture, and Entertainment	0.427
Healthcare	0.256
Other Goods and Services	0.190

Source: Calculated by the author

the gross output of industry n within consumption category i, measured in yuan; N is the total number of industries covered in consumption category i; and the subscript i (i = 1, 2, …, 8) indexes the consumption category, corresponding sequentially to food and alcohol, clothing, residence, household facilities and services, transportation and communication, education, culture and entertainment, health care, and miscellaneous goods and services.

While Equation (2) derives sectoral carbon intensity from production-side industrial data, Equation (3) aggregates consumption-side emissions by coupling these intensities with observed household spending patterns. Specifically, CE represents total household carbon emissions, measured in kilograms of carbon dioxide ($kgCO_2$); CS denotes household consumption expenditure on category i, measured in yuan; and CI is the carbon intensity of consumption category i calculated from Equation (2), measured in kilograms of carbon dioxide per yuan ($kgCO_2$/yuan).

Finally, the carbon emission factors corresponding to per-unit consumption expenditure for each of the eight categories are calculated as shown in Table 2.

3.3. Main variables

3.3.1. Explained variable

The explanatory variable is per capita carbon emissions (ln*($percarbon_{it}$)*), calculated using household carbon emissions divided by household size. The calculation of household carbon emissions includes two approaches: the first is the direct calculation method, which uses the household's energy consumption combined with the carbon emission factor; the second is the indirect calculation method, which uses household consumption data to calculate the carbon emissions behind the consumption. Given the availability of data, this paper uses the second method (Input-Output Method).

3.3.2. Core explanatory variables

The core explanatory variable is the household singlehood rate ($singlehood_{it}$), defined as the proportion of single individuals within a household relative to the total household size. A single individual is identified as any household member aged 20 years or older whose marital status is unmarried, divorced, or widowed.

3.3.3. Control variables

Control variables are selected based on factors known to influence household consumption and, consequently, household carbon emissions. These are organized into two levels: household head characteristics and household-level characteristics.

Household Head-Level Variables:

Age of the household head (and its quadratic term).

Urban/rural residence (1 = urban, 0 = rural).

Household registration type (0 = urban, 1 = rural).

Political affiliation (1 = member of the Communist Party of China, 0 = otherwise).

Gender (1 = male, 0 = female).

Educational attainment (coded as: 0 = illiteracy, 1 = elementary school, 2 = junior high school, 3 = senior high school, 4 = university or above).

Employment status (1 = employed, 0 = unemployed).

Type of employment (1 = works within the system, e.g., government, institutions, state-owned enterprises; 0 = works outside the system).

Self-employment status (1 = self-employed, 0 = otherwise).

Health status (1 = healthy, 0 = otherwise).

Household-Level Variables:

Household size.

Old-age dependency ratio.

Child dependency ratio.

Logarithm of household income.

Logarithm of household net worth.

3.4. Data sources

Data are derived from the China Family Panel Studies (CFPS), a nationally representative longitudinal survey conducted by the Institute of Social Science Survey (ISSS) of Peking University. The survey covers 25 provinces/municipalities/autonomous regions (excluding Xinjiang, Tibet, Qinghai, Ningxia, Inner Mongolia, Hainan, Hong Kong, Macao, and Taiwan), with a target sample of 16,000 households. The CFPS tracks all household members from the baseline year (2010) onward, including subsequent blood and adopted children. The analysis utilizes seven waves of data collected between 2010 and 2022.The data are from China Family Panel Studies (CFPS), funded by PekingUniversity and the Nationl Natural Science Foundation of China. The CFPS is maintained by the Institute of Social Science Survey of Peking University.

To address heteroskedasticity, logarithmic transformations were applied to continuous variables, while proportional variables were retained in their original form. Extreme values in the main variables were winsorized at the 1st and 99th percentiles. Observations beyond three standard deviations of the main variables were excluded to mitigate the influence of outliers and non-normality. The final unbalanced panel dataset comprises 62,412 households.

3.5. Descriptive statistics

Descriptive statistics indicate that annual per capita carbon emissions range from 4.75 kg to 642,204.66 kg, with a mean of 8415.99 kg—consistent with macro-level estimates (see Table 3). The substantial standard deviation (12,798.08) justifies the use of logarithmic transformation to reduce heteroskedasticity. The household singlehood rate varies from 0 to 1, with a mean of 18 % and a standard deviation of 0.18, aligning with figures reported in the Statistical Yearbook.

3.6. Feature fact analysis

As illustrated inFig. 1 and Table 4, China's singlehood rate exhibits a positive correlation with per capita carbon emissions, the latter being calculated based on per capita consumption expenditure. According to data from the China Statistical Yearbook, between 2001 and 2023, annual per capita carbon emissions increased from 1.532 tons to 14.243 tons, while the singlehood rate rose from 25.9 % to 28.8 %.

4. Results

4.1. Baseline regression

The regression results indicate that the singlehood rate has a statistically significant positive effect on per capita carbon emissions at the 1% level (see Table 5). Specifically, a 1 % increase in the overall singlehood rate is associated with a 24.4 % rise in per capita carbon emissions.

The study found that the singlehood rate has a positive effect on per capita carbon emissions. This significant phenomenon stems from multiple underlying factors, including family consumption patterns, resource efficiency, and demographic shifts (Shao et al., 2024) (Cao et al., 2024). The phenomenon is primarily driven by interconnected mechanisms, particularly the reduced resource-sharing efficiency and distinctive consumption patterns characteristic of single-person households.

A key mechanism lies in the fact that single-person households are less efficient than multi-person households in sharing resource consumption. For example, household appliances such as refrigerators, washing machines, and ovens show no significant difference in energy consumption per user within a certain usage range. The per capita energy consumption of a single individual using these appliances is much higher than the per capita energy consumption allocated among two or more people in a household (Jack and Ivanova, 2021). Studies have

Table 3
Descriptive statistics.

VarName	Definition	Obs	Mean	SD	Min	Max
pcarbon	carbon emissions per capita	63412	8415.99	12798.075	4.75	642204.66
singlerate	singlehood rate	63412	0.18	0.315	0.00	1.00
unmarryrate	unmarried rate	63412	0.10	0.232	0.00	1.00
divorcerate	divorce rate	63412	0.02	0.125	0.00	1.00
widowrate	widow rate	63412	0.06	0.198	0.00	1.00
age	age	63412	49.18	14.332	13.00	99.00
age2	square of age	63412	2623.73	1459.121	169.00	9801.00
hukou	registered permanent residence	63412	0.70	0.459	0.00	1.00
urban	urban and rural	63412	0.51	0.500	0.00	1.00
gender	sexual distinction	63412	0.56	0.496	0.00	1.00
party	whether he is a member of the Communist Party	63412	0.04	0.194	0.00	1.00
eduyear	Years of education	63412	7.60	4.912	0.00	23.00
work	whether there is work	63412	0.75	0.432	0.00	1.00
tizhi	whether it is within the country system	63412	0.10	0.306	0.00	1.00
zigu	whether self-employed	63412	0.09	0.283	0.00	1.00
health	health condition	63412	0.36	0.480	0.00	1.00
yanglao	whether there is pension insurance	63412	0.15	0.355	0.00	1.00
yiliao	whether there is health insurance	63412	0.22	0.411	0.00	1.00
gongshang	whether there is work-related injury insurance	63412	0.08	0.273	0.00	1.00
shengyu	whether there is maternity insurance	63412	0.06	0.242	0.00	1.00
shiye	whether there is unemployment insurance	63412	0.07	0.262	0.00	1.00
gongjijin	whether there is a housing fund	63412	0.08	0.276	0.00	1.00
oldrate	The proportion of older persons	63412	0.24	0.364	0.00	1.00
childrate	The proportion of children's	63412	0.07	0.157	0.00	1.00
lnfincome	Household net income	63412	10.63	1.173	0.00	16.59
lntotal_asset	Household net asset	63412	12.21	1.560	0.00	18.20

Source: Calculated by STATA18

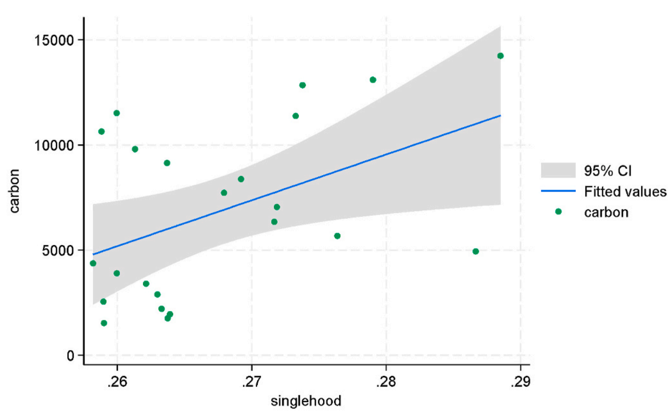

Fig. 1. Singlehood rate and per capita carbon emissions.
Source: China Statistical Yearbook

Table 4
Singlehood rate and per capita carbon emissions(China Statistical Yearbook).

year	pcarbon	singlerate	unmarryrate	divorcerate	widowrate
2001	1531.886	0.259	0.191	0.010	0.058
2002	1761.592	0.264	0.195	0.010	0.059
2003	1952.352	0.264	0.196	0.011	0.057
2004	2209.843	0.263	0.195	0.011	0.057
2005	2551.355	0.259	0.192	0.010	0.057
2006	2895.396	0.263	0.194	0.010	0.059
2007	3403.773	0.262	0.192	0.011	0.059
2008	3899.865	0.260	0.189	0.011	0.059
2009	4375.052	0.258	0.188	0.012	0.058
2010	4942.426	0.287	0.216	0.014	0.057
2011	5676.300	0.276	0.208	0.014	0.055
2012	6347.139	0.272	0.204	0.014	0.053
2013	7052.079	0.272	0.202	0.016	0.054
2014	7728.951	0.268	0.197	0.017	0.054
2015	8380.277	0.269	0.197	0.017	0.055
2016	9148.581	0.264	0.189	0.019	0.056
2017	9804.424	0.261	0.186	0.020	0.055
2018	10643.545	0.259	0.182	0.021	0.056
2019	11518.753	0.260	0.180	0.023	0.057
2020	11382.299	0.273	0.192	0.024	0.057
2021	12848.127	0.274	0.194	0.024	0.057
2022	13102.142	0.279	0.197	0.024	0.057
2023	14242.859	0.288	0.199	0.027	0.063

Source: China Statistical Yearbook

explicitly emphasized that the reduction in household size (often associated with rising single-person rates) poses a significant sustainability challenge by decreasing resource sharing and increasing individual resource consumption. Generally, larger households exhibit lower per capita energy consumption and carbon emissions. For instance, empirical research in China shows that the downsizing of households affects household carbon emissions, indicating that smaller household sizes tend to increase per capita carbon emissions (Yu et al., 2018).

Beyond the use of fixed household appliances, heating and cooling also constitute a significant energy demand. A single-person household requires heating or cooling for their entire living space, resulting in higher per capita energy expenditure compared to multi-person households sharing these costs (Jack and Ivanova, 2021). Similarly, single-person households may have lower food preparation efficiency, often involving small portions that still require full-sized cooking utensils. This leads to higher energy intensity per meal compared to preparing large portions for multiple people.

Travel patterns also exacerbate this effect. Single individuals are more likely to travel alone, leading to lower vehicle occupancy rates and, consequently, higher per capita transportation-related carbon emissions. Although options like carpooling or public transportation exist, daily commutes and personal matters often result in single-person bicycle travel. China's research on household size and transportation carbon emissions confirms that smaller household sizes lead to reduced sharing of transportation resources, increased individual resource consumption, and higher carbon emissions. Considering that transportation is a significant contributor to the overall carbon footprint, this is a key factor (Liang et al., 2024).

Furthermore, the unique consumption patterns of single-person households also contribute to the issue. Individuals living alone may exhibit distinct purchasing behaviors, such as consuming more convenience foods, packaged goods, or dining out more frequently. These

Table 5
Baseline regression.

VARIABLES	(1)	(2)	(3)	(4)
	single	unmarried	divorce	widow
	lnpcarbon	lnpcarbon	lnpcarbon	lnpcarbon
singlerate	0.244***			
	(0.013)			
unmarryrate		0.323***		
		(0.018)		
divorcerate			0.309***	
			(0.031)	
widowrate				0.065***
				(0.022)
age	−0.029***	−0.029***	−0.033***	−0.032***
	(0.002)	(0.002)	(0.002)	(0.002)
age2	0.000***	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
hukou	−0.179***	−0.182***	−0.177***	−0.181***
	(0.016)	(0.015)	(0.015)	(0.015)
urban	0.123***	0.126***	0.124***	0.125***
	(0.016)	(0.016)	(0.016)	(0.016)
gender	0.001	−0.010	−0.004	−0.000
	(0.008)	(0.008)	(0.008)	(0.008)
party	0.143***	0.139***	0.138***	0.138***
	(0.014)	(0.014)	(0.014)	(0.014)
eduyear	0.019***	0.019***	0.019***	0.019***
	(0.001)	(0.001)	(0.001)	(0.001)
work	−0.063***	−0.067***	−0.063***	−0.063***
	(0.010)	(0.010)	(0.010)	(0.010)
tizhi	0.076***	0.080***	0.074***	0.074***
	(0.013)	(0.013)	(0.014)	(0.014)
zigu	0.218***	0.222***	0.219***	0.220***
	(0.014)	(0.014)	(0.014)	(0.014)
health	−0.035***	−0.034***	−0.035***	−0.035***
	(0.008)	(0.008)	(0.008)	(0.008)
yanglao	0.038***	0.039***	0.041***	0.041***
	(0.014)	(0.014)	(0.014)	(0.014)
yiliao	−0.003	−0.001	−0.003	−0.003
	(0.018)	(0.018)	(0.018)	(0.018)
gongshang	0.103***	0.107***	0.111***	0.107***
	(0.019)	(0.020)	(0.020)	(0.020)
shengyu	−0.045*	−0.055**	−0.039	−0.042*
	(0.024)	(0.024)	(0.024)	(0.025)
shiye	−0.070***	−0.067***	−0.070***	−0.066***
	(0.023)	(0.024)	(0.023)	(0.024)
gongjijin	0.069***	0.070***	0.073***	0.072***
	(0.015)	(0.015)	(0.015)	(0.016)
oldrate	−0.067***	−0.037**	−0.069***	−0.078***
	(0.017)	(0.017)	(0.017)	(0.017)
childrate	−0.041*	−0.010	−0.128***	−0.118***
	(0.023)	(0.023)	(0.024)	(0.024)
lnfincome	0.158***	0.152***	0.153***	0.152***
	(0.006)	(0.006)	(0.006)	(0.006)
lntotal_asset	0.072***	0.071***	0.067***	0.065***
	(0.004)	(0.004)	(0.004)	(0.004)
Constant	6.729***	6.800***	6.991***	6.999***
	(0.086)	(0.086)	(0.088)	(0.088)
Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	63,412	63,412	63,412	63,412
R-squared	0.549	0.549	0.546	0.545

Source: Calculated by STATA18

habits often involve higher implicit carbon footprints due to packaging, processing, and transportation processes associated with single-serving meals or restaurant operations. These consumption choices amplify the overall carbon footprint, as household consumption patterns are recognized as direct and indirect drivers of environmental impacts (Duarte et al., 2021)·

The policy implications of these findings suggest that carbon reduction strategies should consider household structures and adopt measures tailored to the consumption patterns of single-person households. This may involve promoting energy-efficient appliances suitable for small families, encouraging resource sharing within communities, or designing urban environments that facilitate sustainable transportation choices (Oltulular, 2025).Carbon footprint analysis tools can help individuals reflect on their environmental impact by considering their socioeconomic background (Brandenstein et al., 2025; Haynes, 2024). Moreover, given the global trend of rising per capita CO2 emissions (Yang et al., 2023), understanding household size-related driving factors is crucial for achieving carbon neutrality goals (Zhang et al., 2023b). The impact of demographic factors on carbon emissions remains a key global concern, while the rising single-person rate further complicates efforts to reduce overall carbon footprints.

Further disaggregation by marital status reveals distinct effects:

The unmarried rate exhibits the strongest positive impact, with a 1 % increase leading to a 32.3 % increase in per capita carbon emissions.

The divorce rate also shows a significant positive effect, resulting in a 30.9 % increase in per capita emissions per 1 % rise.

In contrast, the widowed rate has a comparatively modest effect, with a 1 % increase associated with a 6.5 % increase in per capita carbon emissions.

As for the phenomenon that unmarried people have higher emissions than divorced or widowed people, this difference is mainly due to the combined effect of lifestyle, income level and urbanization.

From the perspective of lifestyle and consumption patterns, unmarried adults (specifically referring to single adults who have never married) typically enjoy higher disposable incomes and greater time flexibility. Their consumption behaviors tend to be more hedonistic, characterized by frequent participation in social dining, travel vacations, and the pursuit of trendy electronic products – activities with high carbon intensity. Additionally, as they mostly live alone, they cannot benefit from the "per capita emission reduction" effect of small-scale families like divorced or widowed groups (who may form multi-member households) through sharing housing, heating, and household appliances with children.

From the perspective of income level, unmarried people are often in the rising stage of their career and have no direct burden of raising children. Their higher disposable income and marginal propensity to consume make them more likely to engage in non-essential consumption with high carbon emissions, while divorced or widowed groups may be more concentrated in necessities due to economic pressure such as property division and single-parent raising.

In addition, in terms of urbanization, unmarried people are highly concentrated in metropolises, and their lives are deeply embedded in the urban infrastructure of high-carbon intensive service systems such as takeout, express delivery and ride-hailing.

Therefore, the higher emissions of unmarried people are essentially a concentrated reflection of their socio-economic characteristics of "more

Table 6
Replace the dependent variable.

VARIABLES	(1)	(2)	(3)	(4)
	singlehood	unmarried	divorce	widow
	lnpcarbon0	lnpcarbon0	lnpcarbon0	lnpcarbon0
singlerate	0.168***			
	(0.024)			
unmarryrate		0.075*		
		(0.044)		
divorcerate			0.336***	
			(0.058)	
widowrate				0.170***
				(0.033)
Constant	4.388***	4.486***	4.461***	4.436***
	(0.137)	(0.138)	(0.139)	(0.139)
Control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	18,647	18,647	18,647	18,647
R-squared	0.426	0.424	0.425	0.425

Source: Calculated by STATA18

individualized lifestyle, higher income and more urbanization".

4.2. Robustness analysis

4.2.1. Replacement of explained variables

Per capita carbon emissions from electricity (lnpcarbon0) were calculated based on household electricity consumption data. The benchmark regression results remain robust. The impact coefficient of the singlehood rate on per capita carbon emissions from electricity is 16.8 % （ see Table 6 ）, which is lower than the benchmark regression coefficient of 24.4 %. This indicates that using electricity consumption alone provides an incomplete representation of energy usage, as it ignores other energy sources and thus leads to an underestimation of the effect of the singlehood rate on carbon emissions.

4.2.2. Replacement of core explanatory variables and removal of some samples

To examine the robustness of the findings, two treatment variables were constructed based on the household singlehood rate. The variable treat0 was defined as 1 if the singlehood rate was greater than 0, and 0 otherwise. Regression results presented in Column 1 （ see Table 7 ） indicate that treat0 has a statistically significant positive effect on per capita carbon emissions, confirming the stability of the baseline results. Similarly, treat1 was defined as 1 only when the singlehood rate equaled 1 (indicating a fully single household), and 0 otherwise. As shown in Column 2 （ see Table 7 ）, treat1 also exhibits a significantly positive impact on per capita carbon emissions, further supporting the robustness of the estimates.

Notably, the coefficient of treat1 is larger than that of treat0. This discrepancy can be attributed to differences in group composition between the two treatment variables. Specifically, treat0 includes all households with any degree of singlehood (from partially single, e.g., singlehood rate = 0.1, to fully single, i.e., singlehood rate = 1), thereby introducing internal heterogeneity. The average treatment effect in this group is diluted by households with low singlehood rates, resulting in a smaller coefficient. In contrast, treat1 comprises exclusively fully single households, which represent a more homogeneous group characterized by stronger carbon-intensive lifestyles—such as greater reliance on takeout meals, higher usage of energy-intensive electrical equipments instead of manual labor, and preference for private transportation. These behaviors lead to higher per capita carbon emissions, which is reflected in the larger coefficient for treat1.

After excluding COVID-19 samples, the regression results remain robust （ Table 7 ）.

Table 7
Replace the core explanatory variables and remove the COVID-19 sample.

VARIABLES	(1)	(2)	(3)
	The singlehood rate is the treatment variable 1	The singlehood rate is the treatment variable 2	Remove the COVID-19 sample
	lnpcarbon	lnpcarbon	lnpcarbon
treat0	0.019** (0.008)		
treat1		0.414*** (0.016)	
singlerate			0.188*** (0.014)
Constant	7.008*** (0.090)	6.416*** (0.089)	6.570*** (0.100)
Control	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	63,415	63,415	48,348
R-squared	0.545	0.555	0.523

Source: Calculated by STATA18

Table 8
Endogeneity analysis.

	(1)	(2)
	stage I	stage II
	Singlehood rate	lnpcarbon
County average singlehood rate	0.940 *** (0.011)	
singlerate		0.306 *** (0.037)
Control	Yes	Yes
Year FE	Yes	Yes
County FE	Yes	Yes
Obs	63415	63415
R2		0.2274
F	7953.11	243.87
CD Wald F	6431.774	
Kleibergen-Paap rk LM 的 p值	0.0000	
Endogeneity test	0.0558	

Source: Calculated by STATA18

4.3. Endogeneity analysis

To mitigate potential endogeneity arising from reverse causality, the average singlehood rate at the county and district levels was employed as an instrumental variable (IV) for the household-level singlehood rate. The two-stage least squares (2SLS) regression results remain statistically significant. The absolute value of the coefficient for the impact of household singlehood rate on per capita carbon emissions is 0.306, which is larger than the baseline regression estimate of 0.244 （ see Table 8 ）.

The first-stage F-statistic is 7953.11, exceeding the conventional threshold of 10, thus satisfying the relevance condition for the instrumental variable. The Kleibergen-Paap rk LM test statistic ($p = 0.0000$) confirms that the equation is identified. The Cragg-Donald Wald F-statistic is 6431.774, far above the 10 % maximal IV size critical value, indicating the absence of weak instrument problems. The endogeneity test yields a p-value of 0.0558, which is less than 0.1, suggesting the presence of endogeneity in the core explanatory variable. variables.

4.4. Mechanism analysis

4.4.1. Economies of scale

Traditional households benefit from stronger economies of scale in the use of household utilities, where high-value assets such as cars and housing are more likely to be shared among family members. In contrast, singlehood households exhibit weaker economies of scale due to a higher likelihood of living alone, reducing the potential for resource sharing. Regression results indicate that the singlehood rate exerts a significant positive influence on per capita housing area, per capita number of housing units, per capita number of cars, and per capita expenditure on car purchases（ see Table 9 ）. Subsequent incorporation of these four mediating variables into the regression models reveals that each has a significant positive effect on per capita carbon emissions, confirming the presence of a mediating mechanism. Notably, after controlling for the number of housing units per capita, the effect of singlehood on per capita carbon emissions becomes statistically insignificant, suggesting that per capita housing units fully mediate this relationship.

4.4.2. Convenience consumption

The singlehood group demonstrates a marked preference for eating out due to the low cost-effectiveness of cooking individually. This group also exhibits higher per capita expenditure on furniture and durable goods, resulting from insufficient sharing of such items within the household. Furthermore, singles show a stronger tendency toward online shopping with home delivery rather than visiting physical stores.

Table 9
The scale economy mediation path of high value public goods in households.

VARIABLES	(1)		(2)		(3)		(4)	
	per-capita housing area		number of housing units per capita		number of cars per capita		per capita car purchase cost	
	phousemianji	lnpcarbon	phousesum	lnpcarbon	pcarsum	lnpcarbon	pcarfei	lnpcarbon
phousemianji		0.253***						
		(0.011)						
singlerate	0.352***	0.160***	0.427***	−0.011	1.509***	0.167***	0.609***	0.249***
	(0.017)	(0.017)	(0.024)	(0.029)	(0.040)	(0.021)	(0.067)	(0.025)
phousesum				0.653***				
				(0.030)				
pcarsum						0.133***		
						(0.006)		
pcarfei								0.230***
								(0.003)
Constant	2.491***	5.945***	1.878***	4.607***	8.261***	5.396***	6.402***	5.768***
	(0.093)	(0.103)	(0.102)	(0.189)	(0.186)	(0.111)	(0.302)	(0.137)
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	27,625	25,782	12,363	11,530	28,575	26,505	13,849	13,190
R-squared	0.880	0.657	0.323	0.602	0.546	0.524	0.131	0.639

Source: Calculated by STATA18

Table 10
Convenience consumption intermediary path of single groups.

VARIABLES	(1)		(2)		(3)		(4)	
	Per capita eat out		Per capita furniture durable goods		Per capita online shopping		Per capita mobile phone fees	
	eat out	lnpcarbon	furniture durable goods	lnpcarbon	online shopping	lnpcarbon	mobile phone fees	lnpcarbon
eat out		0.332***						
		(0.006)						
singlerate	0.605***	0.182***	0.334***	0.174***	0.617***	0.268***	0.169***	0.167***
	(0.026)	(0.015)	(0.040)	(0.017)	(0.045)	(0.026)	(0.016)	(0.015)
furniture durable goods				0.191***				
				(0.003)				
online shopping						0.074***		
						(0.005)		
mobile phone fees								0.280***
								(0.007)
Constant	7.188***	4.745***	5.105***	6.298***	3.130***	6.040***	2.693***	6.038***
	(0.149)	(0.105)	(0.203)	(0.108)	(0.269)	(0.160)	(0.101)	(0.107)
control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,367	21,298	22,544	21,097	13,798	13,057	30,950	29,372
R-squared	0.363	0.594	0.160	0.626	0.354	0.476	0.312	0.498

Source: Calculated by STATA18

Additionally, they allocate more resources to mobile phone services to support remote social interactions, as opposed to in-person family socialization. Accordingly, the following variables were selected as mediators to reflect the singlehood group's pursuit of "convenience-oriented consumption": per capita dining-out expenditure, per capita spending on furniture and durables, per capita online shopping expenditure, and per capita mobile phone expenses. The mediating effects of these variables were all found to be statistically significant (see Table 10) .

4.4.3. *Energy sharing*

The lower degree of energy sharing among the singlehood group results in higher per capita energy consumption, which in turn leads to increased per capita carbon emissions. Household energy consumption is primarily attributed to cooking, heating, and appliance use. Using three types of per capita energy costs—fuel, heating, and electricity—as mediating variables, a chained mediating effect is identified: the singlehood rate influences per capita energy consumption, which subsequently affects per capita carbon emissions (see Table 11) .

4.5. *Heterogeneity analysis*

4.5.1. *Distinguishing between gender and urban/rural*

Gender and urbanization influence household carbon intensity through differentiated lifestyle patterns. The number of male singlehood individuals is defined as those meeting or exceeding the legal marriage age for males (22 years or older), while female singlehood is similarly defined based on the legal marriage age for females (20 years or older). The urban singlehood rate refers to the proportion of single individuals residing in urban households, and the rural singlehood rate denotes that in rural households.

Regression results show that the positive coefficient of the male singleness rate (proportion of single males aged 22 and above relative to total household population) on per capita carbon emissions is significantly larger than that of the female singleness rate (proportion of single females aged 20 and above relative to total household population) (see Table 12). This disparity may be attributable to men's stronger preferences for carbon-intensive lifestyles, such as greater reliance on private car travel, higher meat consumption, alcohol, and tobacco use.

Table 11
Mediating paths of household energy sharing.

VARIABLES	(1)		(3)		(5)	
	Per capita fuel costs		Per capita heating costs		Per capita electricity costs	
	fuel	lnpcarbon	heating	lnpcarbon	electricity	lnpcarbon
fuel		0.272***				
		(0.008)				
singlerate	0.255***	0.192***	0.358***	0.201***	0.237***	0.163***
	(0.018)	(0.015)	(0.034)	(0.033)	(0.015)	(0.012)
heating				0.331***		
				(0.016)		
electricity						0.404***
						(0.007)
Constant	6.541***	5.154***	6.361***	4.866***	5.534***	4.726***
	(0.122)	(0.099)	(0.209)	(0.178)	(0.098)	(0.077)
control	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	37,812	35,685	10,169	9651	51,908	48,985
R-squared	0.167	0.529	0.536	0.516	0.331	0.575

Source: Calculated by STATA18

Table 12
Analysis of gender and urban-rural heterogeneity.

VARIABLES	(1)	(2)	(3)	(4)
	Male singlehood rate	Female singlehood rate	Rural singlehood rate	Urban singlehood rate
	lnpcarbon	lnpcarbon	lnpcarbon	lnpcarbon
singlemanrate	0.275***			
	(0.019)			
singlewomanrate		0.156***		
		(0.017)		
singleruralrate			0.153***	
			(0.013)	
singlecityrate				0.121***
				(0.013)
Constant	6.857***	6.939***	6.914***	6.962***
	(0.087)	(0.087)	(0.087)	(0.088)
control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	63,412	63,412	63,412	63,412
R-squared	0.548	0.546	0.547	0.546

Source: Calculated by STATA18

Table 13
Heterogeneity analysis by age group.

VARIABLES	(1)	(2)	(3)
	Youth singlehood rate	Middle-aged singlehood rate	Older singlehood rate
	lnpcarbon	lnpcarbon	lnpcarbon
singleyoungrate	0.308***		
	(0.019)		
singlemiddlerate		0.227***	
		(0.030)	
singleoldrate			0.120***
			(0.022)
Constant	6.860***	6.985***	6.976***
	(0.086)	(0.088)	(0.089)
control	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Observations	63,412	63,412	63,412
R-squared	0.548	0.545	0.545

Source: Calculated by STATA18

Furthermore, the rural singlehood rate exhibits a stronger positive impact on per capita carbon emissions compared to the urban singlehood rate. This may be explained by the prevalence of high-emission energy practices in rural areas—including the use of coal and firewood for cooking and heating—coupled with less developed infrastructure for public transportation and sharing economy mechanisms relative to urban settings.

4.5.2. Distinguish between ages

Carbon emissions exhibit life-cycle effects, and different age groups contribute differently to these emissions. The singlehood population is categorized into three age groups: young (20–45 years), middle-aged (45–60 years), and elderly (over 60 years). From the age perspective, the positive effect coefficient of the young singleness rate (young single adults as a proportion of total household members) on per capita carbon emissions is larger than that of the middle-aged singleness rate (middle-aged singles as a proportion of total household members), which in turn exceeds that of the elderly singleness rate (elderly singles as a proportion of total household members) (see Table 13). This pattern is consistent with the general observation that aging is associated with reduced carbon emissions.

4.5.3. Distinguishing between incomes

Income level shapes lifestyle patterns, which in turn influence household carbon emissions. Regression analysis based on quartiles of net per capita household income reveals that the singlehood rate exerts a statistically significant effect on per capita carbon emissions in the lowest 25 % and highest 25 % income households, whereas the effect is not significant in the two middle-income quartiles (see Table 14) .

Table 14
Analysis of heterogeneity of income groups.

VARIABLES	(1)	(2)	(3)	(4)
	Lowest 25 %	Below the average 25 %	Above the average 25 %	Highest 25 %
	lnpcarbon	lnpcarbon	lnpcarbon	lnpcarbon
singlerate	0.141***	0.022	−0.019	0.245***
	(0.028)	(0.023)	(0.020)	(0.021)
Constant	7.557***	9.565***	10.627***	8.421***
	(0.152)	(0.159)	(0.173)	(0.203)
control	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Observations	11,346	13,083	13,433	13,771
R-squared	0.211	0.295	0.395	0.440

Source: Calculated by STATA18

Table 15
Analysis of educational heterogeneity.

VARIABLES	(1)	(2)
	High educational singlehood rate	Low educational singlehood rate
	lnpcarbon	lnpcarbon
singlehighedurate	0.352***	
	(0.022)	
singlelowedurate		0.176***
		(0.012)
Constant	6.934***	6.841***
	(0.087)	(0.087)
control	Yes	Yes
Year FE	Yes	Yes
County FE	Yes	Yes
Observations	63,412	63,412
R-squared	0.547	0.547

Source: Calculated by STATA18

This divergence may be attributed to distinct lifestyle patterns prevalent in the extreme income groups. For instance, single individuals in the lowest income group may exhibit limited overall consumption yet maintain carbon-intensive habits—such as smoking, alcohol consumption, and reliance on high-emission fuels like coal. Conversely, those in the highest income group may adopt less carbon-intensive behaviors per unit of consumption, but their high levels of overall consumption—including ownership of luxury cars and larger residences—implicitly result in elevated per capita carbon emissions.

4.5.4. Distinguishing educational qualifications

Individuals with higher education are classified as the highly educated group, while those without are categorized as the low education group. The singles are correspondingly divided into highly educated singles and low education singles. The positive effect coefficient of the high-education singleness rate (proportion of highly educated single members in total household population) on per capita carbon emissions is approximately twice as large as that of the low-education singleness rate (proportion of less-educated single members in total household population) (see Table 15). Although conventional wisdom suggests that highly educated individuals tend to demonstrate a stronger awareness of low-carbon lifestyles, the observed disparity may be attributed to their generally higher income levels and elevated consumption patterns, which could outweigh environmental consciousness in driving carbon emissions.

5. Conclusions and discussion

5.1. Conclusions

This study demonstrates that the household singlehood rate exerts a positive effect on per capita carbon emissions, with an impact coefficient of 0.244. Comparison of influence coefficients reveals that the unmarried rate has a greater effect than the divorce rate, which in turn exceeds the widowed rate. These findings remain robust after replacing explanatory variables, core explanatory variables, and excluding pandemic-affected samples. The mechanisms through which the household singlehood rate affects per capita carbon emissions include: reduced economies of scale in high-value public goods within households, insufficient sharing of household energy, and increased per capita consumption driven by the convenience-seeking behaviors of the singlehood group. After decrease, endogeneity through instrumental variables, the impact coefficient increases to 0.306, and the primary conclusions remain statistically significant.

Heterogeneity analyses indicate that: the impact coefficient of the male singlehood rate on per capita carbon emissions is larger than that of the female singlehood rate; the rural singlehood rate has a greater impact than its urban counterpart; the positive effect of youth singlehood rate is larger than that of the middle-aged group, which in turn exceeds that of the elderly group; singlehood rates in both the highest and lowest 25 % income groups significantly affect per capita carbon emissions, with a larger impact observed in the highest income group; and the positive impact coefficient of singlehood among higher education households is greater in absolute terms than that among lower education households.

5.2. Discussion

5.2.1. Research on the extension of relevant literature

The conceptual framework of this study is rooted in Gary Becker (1965)'s pioneering work on the economics of the household. It analyzes the carbon emission issue through the lens of household demographic structure and consumption, representing a cross-disciplinary integration of sociology, economics, and climate science.

Regarding the principal finding, our research aligns with the seminal works of (Jones and Kammen, 2011)and (O'Neill et al., 2010). Jones and colleagues, focusing on U.S. cities, established that declines in household size elevate per capita energy consumption. Similarly, O'Neill et al.'s cross-national analysis empirically demonstrated that efficiency gains struggle to counteract rising total carbon emissions when the average household size falls below three. Our result—that rising singlehood increases per capita carbon emissions through pathways like weakened economies of scale, stimulated convenience consumption, and decreased energy sharing—essentially confirms and deepens this understanding of structural "diseconomies of scale."

However, there are three major differences between this paper and the existing literature:

First, this study enhances the understanding of the relationship between household structure changes and environmental impacts. The empirical results indicate that the singlehood household rate, as an important sociodemographic variable, has a significant and robust positive influence on carbon emissions—even greater than some traditionally emphasized economic factors. This expands the scope of research on drivers of carbon emissions, highlighting the critical role of micro-level household behaviors and social transformation in macro-level environmental issues.

Second, this research identifies three core mechanisms: diminished economies of scale, insufficient energy sharing, and increased convenience-oriented consumption. These provide a systematic theoretical framework for understanding carbon emission behaviors among single individuals. In particular, the mechanism of convenience-driven consumption underscores the potential environmental pressures arising from modern lifestyles, thereby enriching research in environmental behavioral science and social consumption theory.

Third, detailed heterogeneity analysis—considering gender, urban–rural divide, age, income, and education—reveals the complex ways in which singlehood rates influence carbon emissions. These findings challenge simplified narratives on emission drivers and emphasize the need for differentiated mitigation policies and their micro-level foundations. For example, the greater influence of high-income and highly educated singles offers a new perspective on the conventional assumption that "knowledge promotes environmental protection," prompting further academic inquiry into the complex interplay between knowledge, income, consumption behavior, and environmental responsibility.

5.2.2. Practical implications

(1) Individual carbon budgets are used to encourage singlehood household to adopt green and low-carbon consumption patterns

Personal Carbon Budget (PCB) is a concept in the field of climate change policy aimed at promoting sustainable lifestyles by limiting individual carbon emissions. It sets an emissions cap for citizens, encouraging them to examine and alter their consumption behaviors and daily habits to reduce their carbon footprint (Paulus, 2024).

To effectively address the rising per capita emissions associated with single-person households and leverage the positive role of Personal Carbon Budgets, multi-faceted and tailored policy designs are needed:

First, refine Personal Carbon Budget policies for single-person households. When designing personal carbon allowance policies, the specific characteristics of single-person households should be fully considered. For instance, differentiated carbon allowance standards or targeted incentives could be developed for housing, transportation, and food consumption in single-person households, avoiding a one-size-fits-all approach that may place an unfair burden on this group.

Second, promote scientifically sound carbon footprint measurement tools. Community-level initiatives can use carbon footprint measurement tools to help residents reflect on and reduce their personal carbon footprints, ensuring they stay within their carbon budgets. Exceeding the budget could trigger penalties, such as fines or mandatory participation in activities like tree planting.

Third, synergize with existing carbon reduction policies. Personal carbon allowance policies should not operate in isolation but should work in tandem with existing measures such as energy efficiency standards, renewable energy development, and green transportation infrastructure. For example, carbon tax policies, as an important emission reduction tool, can be integrated with personal carbon allowances, using price mechanisms to guide consumers toward low-carbon products and services (Deng et al., 2023).

(2) By leveraging learning curves strategies, we encourage singles to accumulate and apply low-carbon living knowledge in their daily lives.

This study further posits that for single-person households in Chinese cities, the traditional "diseconomies of scale" effect may be moderated by dynamic "learning effects." The highly digitalized urban environment in China provides single individuals with unique advantages for rapidly learning and adapting to low-carbon lifestyles. For instance, they optimize energy use through smart home applications, reduce resource idleness via sharing economy platforms (e.g., carpooling, shared power banks), and engage in the "social learning" and dissemination of low-carbon knowledge through social media. This "learning-by-doing" process implies that the carbon footprint of thesinglehood household is not a fixed high point but rather a dynamic variable that can be continuously optimized through education and technological innovation.

Based on the theories of the "learning curve (Argote and Epple, 1990)" and "learning by doing (Arrow, 1962)," this paper proposes three interconnected systemic policy recommendations to guide and support urban singles in adopting low-carbon lifestyles.

First, we recommend establishing a national-level "Personal Low-Carbon Living Digital Assistant" platform. Led by the government and jointly developed with internet technology companies and environmental organizations, this platform aims to systematize, streamline, and personalize individuals' learning and practice processes. Its core functions should encompass three aspects: First, enabling real-time carbon footprint tracking and precise feedback. By integrating smart meter and transportation data, it would provide users with concrete, actionable suggestions such as "Raising your AC temperature by 1° could save X kWh this week."

Second, incorporating scenario-based learning modules. Through engaging tasks and challenges covering daily scenarios like dining, accommodation, and transportation, users would accumulate knowledge and receive incentives during completion. Third, it should foster low-carbon living communities to facilitate peer-to-peer experience sharing and skill exchange. Algorithms would intelligently push success stories from similar users, amplifying the "social learning" effect. This approach aims to significantly lower the barriers to learning and applying low-carbon knowledge through digital means, accelerating the "learning curve" for single individuals.Building on digital empowerment, the "Community Low-Carbon Living Labs" initiative should be implemented concurrently, extending learning effects from online to offline physical spaces. This initiative can utilize community public spaces to establish small-scale experimental or experiential corners. Specifically, it can include smart home device experience zones where residents can personally operate energy-efficient appliances and tangibly feel the benefits of technology; regularly host skill workshops on upcycling old items and energy-efficient cooking; and set up tool and book sharing corners to encourage resource recycling. Additionally, it should showcase actual low-carbon achievements and energy-saving data from community residents. By collaborating with property management services, localized incentives like discounted fees and priority access to community services can be implemented. This approach bridges the perceptual gap inherent in purely online interactions, deepening understanding through hands-on practice and internalizing low-carbon concepts into positive lifestyle habits rooted in community belonging.

Finally, to sustain learning motivation, establishing a "Green Credit Points" system linked to the aforementioned platform and lab is essential. This system's core principle is transforming dynamic learning effects into lasting incentives. Points are quantified based on multi-dimensional behaviors like energy-saving achievements, knowledge acquisition, and community participation, with diverse redemption channels available. Points can be redeemed for public services like priority access to popular sports facilities or discounts on public parking fees, as well as for benefits in areas such as shared mobility services and e-commerce platforms through corporate partnerships. Furthermore, financial institutions could be encouraged to offer green financial products to high-point users, while the rental market could be guided to provide preferential treatment, thereby tightly linking environmental actions with economic and social benefits. In summary, these three proposals synergistically leverage digital tools, tangible scenarios, and systematic long-term incentives to form a complete policy loop—from awareness guidance to practice conversion to value feedback—systematically helping singles reduce their carbon footprint.

(3) By intervening at sensitive points, we help singlehood household adopt low-carbon lifestyles.

Sensitive Intervention Pointsrefer to situations in complex economic systems where small, targeted policy or social interventions may trigger significant, positive structural changes through amplification effects within the system. The "small interventions" targeting the singlehood households in Chinese cities could lead to "big changes" (Thaler and Sunstein, 2008).

At the community level, make renting items easier.Introducing shared refrigerators, shared laundry rooms, and item exchange corners in communities with high concentrations of single youth (e.g., youth apartments in large cities) can serve as "sensitive intervention points." These low-cost interventions create new models for resource sharing, directly countering "diseconomies of scale."

At the policy level: Integrating digital "personal carbon accounts" with social media and e-commerce platforms, where every low-carbon action (e.g., green travel, paperless offices) by single youth receives immediate feedback and incentives (e.g., coupons, credit points), represents another "sensitive intervention point" leveraging behavioral psychology.

At the market level, we should develop the sharing economy model. By providing long-term rental apartments with public areas, shared cars, gyms, offices, study rooms, and karaoke lounges, we can help singlehood household fully enjoy the dual benefits of cost-effectiveness and low-carbon emissions in their daily lives. In the future, we should particularly promote service sharing, such as offering ready-to-cook meals in cafeterias, which will reduce carbon emissions from individual cooking.

Regarding intervention timing, it's crucial to capitalize on pivotal consumption moments that influence long-term carbon emissions for single-person households. Key opportunities include first-time rentals, appliance purchases, and car acquisitions. The mobile shopping system should proactively recommend low-carbon alternatives like shared apartments, energy-efficient appliances, and electric vehicles, accompanied by discount coupons. When households move, upgrade appliances, or replace vehicles, the system should automatically suggest eco-friendly products. This approach encourages single-person households to prioritize low-carbon purchases and develop sustainable consumption habits.

(4) Building a low-carbon and inclusive society,

When proposing social policies targeting the carbon emissions of single-person households, it is essential to first clarify their ethical boundaries. All policy recommendations in this study adhere to the principles of diversity, equity, and inclusion, as well as the framework of a just transition. The core stance is that the policy objective should not be to alter an individual's marital status but to provide all citizens, regardless of household structure, with the capacity and choices to transition to low-carbon living. We must respect individuals' autonomy in lifestyle choices and avoid stigmatizing single individuals or portraying them as the root of social problems. The responsibility for emission reduction should be shared fairly, and policy efforts should focus on building an inclusive low-carbon support system rather than intervening in specific lifestyle choices.

Building an Inclusive Low-Carbon Support System for All Household Structures:

First of all, promote energy-saving solutions tailored for small households: Develop and subsidize high-efficiency appliances and modular furniture suitable for single apartments and small units, and provide household energy management consulting services tailored to those living alone to help them achieve energy savings at lower costs.

Second, develop community sharing economies to mitigate diseconomies of scale: Encourage and invest in the construction of shared spaces and facilities at the community level, such as public kitchens, shared laundry rooms, tool libraries, and community canteens. This not only effectively reduces per capita resource consumption for those living alone but also fosters community integration without imposing mandatory household consolidation.

Third, design inclusive urban services and infrastructure: Optimize urban planning to increase the supply of affordable small-sized housing, and improve public transportation, walking, and cycling networks to ensure that single residents and all other citizens have convenient access to low-carbon travel options.

Last but not least, implement public education on low-carbon living: Enhance carbon footprint awareness and emission reduction skills among all citizens (not solely targeting single individuals) through public campaigns and educational programs, empowering everyone to make more environmentally friendly consumption decisions.

5.2.3. Limitations and future research

(1) This study has several limitations, including insufficient data granularity, constrained exploration of micro-level behavioral mechanisms due to data limitations, lack of cross-country cultural comparisons, and absence of policy simulation and cost–benefit analysis for the proposed measures.
(2) Data limitations prevent access to specific consumption figures for fossil and non-fossil energy sources, making it difficult to measure direct carbon emissions with greater precision.
(3) Limitations in data privacy protection make it challenging to conduct county-level regional heterogeneity analysis using economic data from various counties and districts.

Based on these findings and limitations, future research should:

(1) Based on various energy consumption data (distinguishing between fossil and non-fossil energy sources), the direct carbon emissions of households with different single-person rates were calculated in detail using the life cycle assessment method;
(2) Refine micro-level behavioral mechanism studies using detailed surveys, experiments, or big data to examine energy use and emission behaviors of singles in specific contexts (e.g., housing, transportation, consumption);
(3) Conduct cross-cultural comparative studies to examine the generalizability of the relationship between singlehood and carbon emissions across different sociocultural settings (e.g., Europe, North America, East Asia), exploring the moderating roles of cultural values and social welfare systems;
(4) Perform policy simulations and cost–benefit analyses to model the environmental and socioeconomic impacts of various interventions (e.g., marriage subsidies, shared housing, carsharing, energy-efficient technology promotion), providing quantitative evidence for policy design.

Institutional review board statement

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Institutional Review Board" Biomedical Ethics Committee of Peking University" (protocol code IRB00001052-14010.

Data availability statement

The data are from China Family Panel Studies (CFPS), funded by Peking University and the National Natural Science Foundation of China. The CFPS is maintained by the Institute of Social Science Survey of Peking University (https://www.isss.pku.edu.cn/cfps/sjzx/gksj/index.htm).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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